KERNEL OLYMPICS

CUDA → ROCm Migration Copilot

Ship AMD-ready code in minutes, not months.

AMD Developer Hackathon ACT II • Track 3 — Unicorn
Team Meteorite □ • Team-3793

github.com/indrad3v4/Kernel-Olympics • endearing-rebirth.up.railway.app

The $10B Problem

80%

hipify handles easily

20%

Manual weeks-long engineering — warp ops, shfl, libraries

$10B/year

Enterprise migration friction cost

2,400+

CUDA kernels per average large codebase

MI300X outperforms NVIDIA on price/performance. Software adoption gap is the blocker.

Enterprises stay on CUDA because the tail 20% takes weeks per project. Kernel Olympics kills that wait.

Solution: Kernel Olympics

A 4-LLM agentic loop that ports, validates, and verifies — no human in the middle.

SCAN
hipify + risk classification

→

PLAN
DeepSeek v4 architectural plan

→

⚡ **PORT**
GLM-5.2 kernel code gen

→

EVAL
Kimi-K2.7 3-gate validation

→

VERIFY
hipcc + MI300X real GPU run

→

REPORT
Gemma summary report

~105s	~$0.03	30+/hr	<1%
Per kernel	LLM cost	Throughput	False positive

Architecture: Multi-Agent Orchestration

DeepSeek v4 — Planner → **GLM-5.2** — Coder → **Kimi-K2.7** — Evaluator → **Gemma** — Fallback Eval

DeepSeek v4	GLM-5.2	Kimi-K2.7	Gemma
Planner	Coder	Evaluator	Fallback Eval
Plans structure	Generates HIP	Validates output	Failsafe

_strip_to_kernel_only

Pre-processor strips host code → coder sees only kernel functions. Eliminates line offset pollution from preamble.

Three-Gate Validation

Lexical (rejects prose) → Structural (braces/symbols) → Semantic (31 tests, <1% FP). Halts bad output before compile.

Pattern Memory

Trigram index + 0.2ms lookup. Cache hits skip LLM entirely — ~60,000× faster than an LLM call.

All LLMs hosted on AMD hardware via Fireworks AI API • Runs on CPU (no GPU needed for pipeline)

Smart Porting: _strip_to_kernel_only

Before (full CUDA source → coder)

```
// Copyright (c) 2024 NVIDIA Corporation
// All rights reserved.

#include <cuda_runtime.h>
#include <cuda.h>

__host__ void init() {
  cudaSetDevice(0);
}

__global__ void kernel() {
  int tid = threadIdx.x;
  __syncthreads();
}

int main() {
  init<<<1,32>>>();
  return 0;
}
```

⚠ 30-line preamble → line offset pollution → false compile errors

After (kernel-only → coder)

```
__global__ void kernel() {
  int tid = threadIdx.x;
  __syncthreads();
}
```

✓ Strips includes, host code, main() — coder sees ONLY kernel bodies

Result: Precise line offsets → reliable HIP output → fewer retries → 2x faster pipeline

Three-Gate Validation System

01
Lexical Gate

- Rejects prose/markdown in code output
- Checks for hallucinated text blocks
- Ensures only valid CUDA/HIP syntax

Pass rate: ~92%

02
Structural Gate

- Braces balance verification
- Symbol table preservation check
- Function signature matching

Pass rate: ~88%

03
Semantic Gate

- 31 test cases for code correctness
- HIP API substitution validation
- <1% false-positive rate

Pass rate: ~96%

Failed at any gate → auto-retry with gate error context → up to 3 retries → graceful failure report

Performance & Speed

~105s
Pipeline per kernel

~$0.03
Average cost

30+
Kernels/hr

vs. Manual Engineering

Approach	Time	Cost	Confidence
Manual (expert engineer)	2–4 weeks	~$8,000	Variable
hipify + manual fix	3–5 days	~$2,500	Medium
KernelOlympics	**105 seconds**	**~$0.03**	**96%**

⚡ Pattern memory: 0.2ms lookup vs ~12s LLM call = 60,000× speedup for cached patterns

Team Meteorite

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AI Architect / Lead

Pipeline architecture, LLM orchestration, 3-gate validation, Railway deployment

Aahil-Riyaz
AMD/ROCm Engineering

hipcc compilation, MI300X verification, GPU testing pipeline

Bromine185
Kernel Engineering

CUDA kernel analysis, HIP translation patterns, unicode/edge cases

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CI/CD & Testing

GitHub Actions, test suite (605+), build automation

Icodemun44
Infrastructure

Containerization, environment setup, dependency management

_dD
QA & Documentation

Debug mode, summary reports, project documentation

THANK YOU

Try Kernel Olympics — it's open source

github.com/indrad3v4/Kernel-Olympics

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Questions?

"Like a meteorite, we don't arrive quietly."